TopoMortar: A Dataset to Evaluate Topology Accuracy in Image Segmentation

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Abstract

We present TopoMortar, a brick wall dataset that is the first dataset specifically designed to evaluate topology-focused image segmentation methods, such as topology loss functions. Motivated by the known sensitivity of methods to dataset challenges, such as small training sets, noisy labels, and out-of-distribution test-set images, TopoMortar is created to enable in two ways investigating methods' effectiveness at improving topology accuracy. First, by eliminating dataset challenges that, as we show, impact the effectiveness of topology loss functions. Second, by allowing to represent different dataset challenges in the same dataset, isolating methods' performance from dataset challenges. TopoMortar includes three types of labels (accurate, pseudo-labels, and noisy labels), two fixed training sets (large and small), and in-distribution and out-of-distribution test-set images. We compared eight loss functions on TopoMortar, and we found that clDice achieved the most topologically accurate segmentations, and that the relative advantageousness of the other loss functions depends on the experimental setting. Additionally, we show that data augmentation and self-distillation can elevate Cross entropy Dice loss to surpass most topology loss functions, and that those simple methods can enhance topology loss functions as well. TopoMortar and our code can be found at https://jmlipman.github.io/TopoMortar.

1 Introduction

Deep learning has demonstrated extraordinary potential for image segmentation, yet even state-of-the-art models [26] cannot guarantee the connectivity of thin tubular structures, such as axons, vessels, and fibers. As a result, minor misclassifications can break the continuity of these structures, compromising their subsequent quantification. Topology loss functions [21] aim to address this issue by encouraging models to produce segmentations with the correct number of topological structures, such as connected components, holes, and hollows. However, their effectiveness is not completely well understood due to limitations in the datasets used to evaluate them.

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Topology loss functions have been evaluated on datasets with regions requiring precise connectivity, such as blood vessels. Datasets, in addition to their high-level segmentation task (*e.g.*, segmenting blood vessels on fundus retina images), present **challenges** that are rarely discussed or accounted for, such as class imbalance, small dataset size, noisy labels, pseudo-labels, and out-of-distribution (OOD) test-set images. The entanglement between the dataset task and dataset challenges obscures understanding when and where topology loss functions improve topology accuracy. For instance, a method addressing the same challenge across different datasets (*e.g.*, Dice loss in class-imbalanced datasets) may increase topology accuracy simply because it improves accuracy by tackling that particular challenge; however, it will not improve topology accuracy in similar datasets with other challenges. Separating dataset task from dataset challenges to investigate methods' **robustness** against such challenges allows elucidating whether methods learn data's topological properties [15]. Additionally, topology loss functions have not yet been evaluated on a dataset without challenges (*e.g.*, class imbalance, imperfect labels)—likely, because such a dataset does not exist. A challenge-free dataset would reduce the possibilities for methods to increase topology accuracy by tackling dataset challenges. Thus, an increase in topology accuracy can be attributed to the method's capability to learn the data's topological properties. It is also unknown whether simple methods addressing dataset challenges, such as data augmentation and self-distillation, are more effective than topology losses, and whether topology loss functions can be further improved with such methods. Our main contributions are:

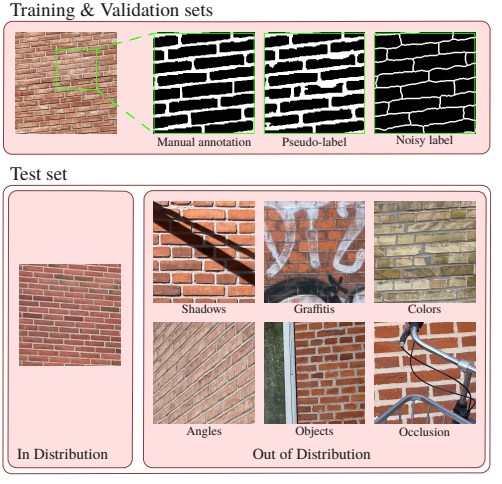

Figure 1: The TopoMortar dataset.

- We release the first dataset specifically acquired to investigate whether topology-focused methods improve topology accuracy and effectively learn data's topological properties.
- We compare extensively, with 10 random seeds and statistical significance tests, Cross entropy Dice loss with six topology and one non-topology loss functions.
- We show that clDice was the only evaluated topology loss function that improved topology accuracy in most of the experiments, demonstrating its effectiveness.
- We demonstrate that data augmentation and self-distillation can increase topology accuracy even when optimizing topology loss functions.

2 Related work

Topology loss functions have been evaluated on many different image segmentation datasets where topology accuracy has been considered essential. The most used datasets across 28

related studies [2, 8, 10, 11, 16, 17, 18, 20, 21, 22, 23, 25, 27, 28, 29, 35, 36, 37, 40, 42, 43, 44, 45, 47, 49, 50, 52, 53] were DRIVE, Massachusetts Roads, and CREMI. DRIVE [46] is a dataset of fundus retina images for blood vessel segmentation that has 20 training and 20 test images. The Massachusetts Roads dataset [34] consists of 1171 aerial images for road segmentation split into 1108, 14, and 49 training, validation, and test set images. The CREMI dataset[1] is comprised of three 3D electron-microscopy images of the brain tissue of adult Drosophila melanogaster. Other datasets used to evaluate loss functions are CrackTree [55] (photographs of concrete cracks); ISBI12 [4] and ISBI13 [5] (electron-microscopy images of neurons); RoadTracer [5] and DeepGlobe [12] (aerial images of roads); and ACDC [7] and left ventricle UK biobank [39] (cardiac magnetic resonance images).

These datasets present different challenges, making it difficult to determine whether improved topology accuracy stems from a method's suitability to dataset challenges, its suitability to a specific task, or its effective learning of the data's topological properties. The DRIVE dataset is extremely *small*; around one-third of the training set images of the Massachusetts Roads dataset are *corrupted* (see Appendix A); The CREMI dataset only provides the instance segmentation of the neurons, thus, each study had to derive its own neuron borders *pseudo-labels*—a process that has not been documented and has been likely carried out differently (see Appendix B); CrackTree's labels are one-pixel width lines (*i.e.*, *noisy labels*, see Appendix C). On DRIVE—the most used dataset—previous studies have conducted 5-fold cross-validation on the 20 training set images [25], 3-fold cross-validation on 30 images [44], 3-fold cross-validation on the 20 training set images [21, 22], applied a 16-4-20 training, validation, and test set split [2, 17, 43], a 16-4 train-test split [29], or an unspecified split on the 20 training set images [20, 40]. This inconsistency is likely due to a combination of factors, including its small size, the lack of a fixed train-validation split, and the unavailability of the test-set labels. More details on the datasets and experimental discrepancies across studies can be found in Appendix D.

3 TopoMortar

Our dataset, TopoMortar, is a brick wall dataset consisting of 420 RGB images of 512 $\times$ 512 pixels for the task of mortar segmentation. We have chosen brick walls with mortar because of the well-defined topological properties of the bricks and mortar. The mortar allows variation in the labels, and the brick walls are well suited for testing specific shifts in domain such as occluding objects, color change, etc. TopoMortar's task (*i.e.*, mortar segmentation in red brick walls) is relatively simple by design; in contrast to existing datasets that are more complex, an improvement in topology accuracy in TopoMortar is less likely to be due to a more suitable choice of the neural network, optimizer, training time, etc.

TopoMortar is built to address previous dataset limitations and to avoid discrepancies in the experimental settings of future studies. To this end, TopoMortar includes 1) a fixed training-validation-test set split (50-20-350), 2) two fixed training sets (50 and 10), 3) accurate, noisy, and pseudo-labels for the training and validation sets, 4) the manual annotations of all the images, and 5) several OOD test set images (85% of the test set) divided into six groups portraying different challenges (see fig. 1 (top)). The training and validation sets contain images that align with the general concept of a red brick wall, *i.e.*, reddish bricks with mortar horizontally and vertically oriented and without any shadows or objects. The test

[1] https://cremi.org

set images are divided into seven groups: *in-distribution* brick walls that are similar to the training and validation sets; brick walls with *shadows* and *graffitis*; brick walls with bricks of different *colors*; brick walls images non-horizontally aligned taken from a different *angle*; and brick walls with *objects* in/next to them and objects *occluding* the walls. Figure 1 (bottom) shows an example of each category. TopoMortar is larger than most datasets used in previous related studies (Appendix D). Moreover, as our experiments demonstrate, its large training set consisting of 50 images suffices to achieve significantly higher topology accuracy than the small training set of only 10 images, allowing to study whether topology losses advantageousness decreases when increasing the training set size.

TopoMortar allows to investigate whether methods are effective at improving topology accuracy in two ways. First, by **eliminating dataset challenges** (*i.e.*, confounding factors such as scarce training data, inaccurate labels, and OOD test-set images) with TopoMortar's large training set, accurate labels, and ID test set. Without dataset challenges, a method has limited ability to exploit specific dataset characteristics to increase accuracy and, therefore, topology accuracy. Thus, an improvement in topology accuracy can be attributed to the effective learning of topological properties. Second, by permitting to assess, on the same dataset, model **robustness against various dataset challenges**: scarce training data, pseudo-labels, noisy labels, and OOD test-set images. By utilizing the same training set images (thus, fixing the dataset-related effects), an increase in topology accuracy across all challenges indicates the learning of topology information. More details about TopoMortar, its labels, and its suitability for assessing topology accuracy can be found in Appendix E.

4 Experiments

We conducted four sets of experiments. First, we investigated the impact of dataset challenges and limitations on the effectiveness of topology loss functions in datasets used by previous work. Second, we compared topology loss functions on TopoMortar in a setup without dataset challenges. Third, we compared topology loss functions across different dataset challenges. Fourth, we studied the extent to which two simple methods for tackling dataset challenges that do not directly aim at increasing topology accuracy (data augmentation and self-distillation) can actually increase topology accuracy.

We compared Cross entropy and Dice loss (CEDice), RegionWise loss [48], TopoLoss [21], TOPO [36], Warping loss [22], clDice [44], Skeleton Recall [25], and cbDice [43]. More details about the loss functions can be found in Appendix F. In addition to computing Dice coefficient [13] and Hausdorff distance (95th percentile) [41], we computed the Betti errors. The Betti 0 error (β_0) refers to the difference in the number of connected components, while the Betti 1 error (β_1) refers to the difference in the number of holes (in most cases corresponding to the bricks in TopoMortar).

All our experiments were run with 10 different random seeds, providing us with sufficient measurements to evaluate the significance of performance differences. For this, we computed paired permutation tests with 10,000 random iterations. We trained nnUNet for 12,000 iterations on batches of 10 images with deep supervision [51] and stochastic gradient descent, with a learning rate of 0.01, nesterov momentum of 0.99, weight decay of 3×10^{-5}, and polynomial learning rate decay $(1 - \frac{iteration}{12000})^{0.9}$. We employed 10 data augmentation transformations (see Appendix G). We implemented our experiments in MONAI [9] and PyTorch [38], and we ran our experiments in two clusters with several Tesla A100, V100, A10, and A40, ranging from 16 to 40 GB of GPU memory.

4.1 Challenges and limitations in previous datasets

CREMI lacks true labels, but for consistency with previous work we also utilized it. We divided CREMI into one image for training, one for validation, and one for testing, and we compared all loss functions, with and without data augmentation. The only loss functions that achieved smaller and significantly different β_1 errors than CEDice were TopoLoss and RWLoss (see Table 1, "CREMI"). However, since CREMI's pseudo-labels had many holes and the Betti errors only focused on their number, automatic segmentations with numerous holes, including incorrect ones, will show lower β_1 errors. In consequence, a decrease in the β_1 error does not guarantee higher topology accuracy; instead, it might indicate that the segmentation had more incorrect holes (see Appendix H), making CREMI unsuitable for quantifying β_1 errors. Data augmentation, which is heavily under-reported in the literature (Appendix D), was crucial to improving accuracy. Furthermore, we observed that it was possible to make any loss function appear as the best by carefully selecting a random seed (Appendix I).

		Betti error		Dice	
		D.A.	No D.A.	D.A.	No D.A.
CREMI	CEDice	2371±1046	1356±1126	0.81±0.0	0.75±0.0
	RWLoss	3008±1008	**1039±1145**	0.79±0.01	0.77±0.0
	TopoLoss	**1898±1090**	3393±1206	0.81±0.0	0.75±0.0
	TOPO	16140±30223	20327±1687	0.63±0.22	0.78±0.0
	clDice	2904±635	6421±1253	0.77±0.0	0.76±0.0
	Warping	3795±998	1270±620	0.80±0.0	0.76±0.0
	SkelRecall	2519±1323	2003±1198	0.75±0.01	0.75±0.0
	cbDice	3407±951	2934±1460	0.81±0.0	0.76±0.0
		DRIVE	FIVES	DRIVE	FIVES
Supervised	CEDice	172.42±15.7	43.79±6.74	0.72±0.0	0.89±0.0
	RWLoss	**159.38±15.4**	163.9±17.78	0.74±0.0	0.76±0.01
	TopoLoss	**67.54±8.1**	**42.51±7.15**	0.73±0.0	0.86±0.0
	TOPO	221.6±37.08	76.83±16.51	0.72±0.01	0.81±0.01
	clDice	**61.04±8.89**	**16.99±3.97**	0.73±0.0	0.86±0.0
	Warping	**115.62±11.38**	53.41±7.11	0.73±0.0	0.88±0.0
	SkelRecall	**128.86±15.28**	**37.75±6.98**	0.72±0.0	0.83±0.01
	cbDice	**116.32±12.02**	55.69±7.77	0.73±0.0	0.87±0.0
		Supervised	Adele	Supervised	Adele
CrackTree	CEDice	84.3±9.66	51.97±7.56	0.78±0.01	0.75±0.01
	RWLoss	-	-	-	-
	TopoLoss	145.82±18.18	134.87±19.59	0.52±0.02	0.47±0.02
	TOPO	-	-	-	-
	clDice	**22.13±28.54**	**18.03±37.17**	0.54±0.12	0.34±0.08
	Warping	94.85±10.71	**46.37±7.35**	0.77±0.01	0.73±0.01
	SkelRecall	**14.03±5.82**	**24.62±8.31**	0.32±0.01	0.3±0.01
	cbDice	84.64±8.33	66.49±9.3	0.76±0.01	0.69±0.01

Table 1: Mean and std. of Betti errors in previous datasets. Top: β_1 error on CREMI dataset, with vs. without data augmentation. Center: β_0 error in standard supervised training, DRIVE vs. FIVES datasets. Bottom: β_0 error on CrackTree dataset, standard supervised learning vs. Adele. **Bold**: Betti errors are lower and significantly different than CEDice loss.

DRIVE is extremely small, obscuring whether topology loss functions are particularly beneficial on fundus retina images or on datasets of small size. To answer this question, we compared topology loss functions on DRIVE (13-2-5 train-validation-test split) alongside FIVES [24] (538-60-200 split), which is a similar but much larger dataset. On the DRIVE dataset, six loss functions achieved smaller and significantly different β_0 errors than CEDice, whereas, on FIVES, only three of them achieved smaller and significantly different β_0 errors (Table 1, "Supervised"), demonstrating the impact of scarce data on topology accuracy. Loss functions generally performed better on the FIVES dataset, with CEDice gaining a nearly $\times 4$ improvement in the β_0 error—the largest. Additionally, while the Dice coefficients were similar across loss functions within the same dataset, the β_0 errors varied considerably. clDice achieved the most topologically accurate segmentations on both datasets, and the relative effectiveness of the other loss functions varied. For instance, Skeleton Recall was the 5$^{\text{th}}$ most accurate loss on DRIVE, but the 2$^{\text{nd}}$ on FIVES.

	Loss	β_0 error	β_1 error	Dice	HD95
ID test set	CEDice	3.31±2.62	2.13±1.02	0.91±0.00	1.87±0.03
	RWLoss	5.72±1.65	6.03±2.15	0.91±0.00	1.84±0.01
	TopoLoss	3.14±1.80	2.73±3.37	0.91±0.0	1.87±0.03
	TOPO	33.69±4.65	43.35±3.44	0.86±0.00	2.68±0.18
	clDice	**1.17±0.54**	**1.41±0.59**	0.91±0.00	1.83±0.01
	Warping	4.20±2.04	3.62±1.16	0.91±0.00	1.84±0.01
	SkelRecall	3.08±1.41	**1.73±0.77**	0.91±0.00	1.90±0.03
	cbDice	4.24±2.08	3.81±2.28	0.91±0.00	1.85±0.02
OOD test set	CEDice	197.25±24.21	87.81±21.32	0.65±0.01	37.65±1.20
	RWLoss	219.55±28.45	87.74±19.22	0.64±0.01	37.31±1.29
	TopoLoss	**193.29±47.92**	99.33±28.05	0.65±0.01	37.33±1.06
	TOPO	**117.73±19.37**	105.09±24.71	0.65±0.01	40.21±0.81
	clDice	**81.33±13.16**	**51.64±7.34**	0.66±0.01	36.90±1.29
	Warping	204.26±26.69	**80.23±18.01**	0.65±0.01	37.07±1.25
	SkelRecall	**180.23±24.44**	96.79±30.14	0.67±0.01	38.46±1.02
	cbDice	233.49±34.09	106.08±33.24	0.65±0.01	37.84±1.16

Table 2: Average performance (10 random seeds) on TopoMortar test set, separated into in-distribution (ID) and out-of-distribution (OOD) images. Training setup: Standard supervised learning, large training set, accurate labels. **Bold**: Betti errors are lower and significantly different than CEDice loss.

CrackTree's noisy labels were annotated with one-pixel width lines (see Appendix C). We compared optimizing the topology loss functions via standard supervised learning, which is suboptimal for this type of labels, and via Adele [51], which is a method designed for training deep learning models with noisy labels. We divided CrackTree into a 147-17-42 train-validation-test split and we tackled class imbalance by multiplying the loss on each class by [0.2, 0.8]. All topology loss functions improved their topology accuracy when optimized via Adele (Table 1, "CrackTree"), except Skeleton Recall loss that even with standard supervised learning it achieved the lowest β_0 errors. RWLoss and TOPO led to empty masks. Since the test-set labels are also noisy, the Dice coefficients hardly reflected segmentation quality. For instance, clDice and Skeleton Recall, which achieved the lowest Dice coefficients, produced thick segmentations that corresponded better to the exact location of the concrete cracks than the ground truth (Appendix C). In general, Adele led all loss functions to produce thicker segmentations, decreasing their Dice coefficients.

4.2 Benchmark on TopoMortar without challenges

We evaluated topology loss functions on TopoMortar on a setup without dataset challenges, thus, **reducing the confounding factors** that could lead to an increase in topology accuracy without learning the data's topological properties. To this end, we trained on TopoMortar's *large* training set with *accurate* labels, and we separated the performance measurements in the test set between *ID* and OOD. In other words, in this experiment, we accounted for no dataset challenges, as prior topology loss function studies, but, differently from those studies, we ensured our dataset had no such challenges, which, as we showed in Section 4.1, affected topology accuracy. On top of the Betti errors, Dice coefficient, and HD95, we also measured local topology accuracy by computing the Betti error in a 128 × 128 sliding window.

clDice and Skeleton Recall were the only loss functions that achieved Betti errors lower and significantly different than CEDice on both the ID and OOD test-set images (see Table 2). In the ID test set, CEDice, TopoLoss, and Skeleton Recall produced segmentations with the second lowest β_0 errors, which, between them, were not significantly different. In the OOD test set, TopoLoss and TOPO also achieved lower and significantly different β_0 errors than CEDice, while Warping did similarly on the β_1 error. TOPO's low β_0 errors in the OOD dataset were due to over-segmentation, especially in the bricks with different colors (see appendix J, "colors"). The Dice coefficients and HD95 were similar across loss func-

tions and, although they did not reflect segmentation quality too accurately, they signaled whether a loss function did not produce satisfactory segmentations (see TOPO in Table 2 (ID test set and Dice, and OOD test set and HD95)). Since the local Betti errors were highly correlated to β_0 and β_1 errors (Pearson correlation > 0.98), we did not include them in the paper.

4.3 Robustness to scarce training data, low-quality labels, and OOD images

We investigated on TopoMortar whether and to what degree existing topology loss functions enhance model **robustness** against scarce training data, inaccurate labels, and OOD images. Studying model robustness by disentangling the different types of dataset challenges and other dataset-related factors allows elucidating if topology loss functions learn data's topological properties [15]. First, we compared topology loss functions in a scarce training data setup (as in DRIVE) by using TopoMortar's *small* training set. Second, we compared them in a setup with labels generated semi- or fully automatically (as in CREMI) by using TopoMortar's *pseudo-labels*. Third, we compared topology loss functions in a setup with inaccurate labels resulting from a quick approximated human annotation (as in CrackTree) by using TopoMortar's *noisy labels*. Additionally, we separate the measurements distinguishing between ID and OOD test-set images. In all the experiments, we trained the models via standard supervised learning and, unless otherwise specified, we employed TopoMortar's large training set and accurate labels.

	Test set →	β_0 error ID	β_0 error OOD	β_1 error ID	β_1 error OOD
Small training set	CEDice	9.6±4.4	182±16	4.6±3.3	61.5±8.6
	RWLoss	12.4±7.2	201±17	6.4±2.8	**56.5±12**
	TopoLoss	**8.7±3.0**	**152±14**	4.8±5.1	**58.1±8.9**
	TOPO	60.0±20	312±34	43.3±3.5	109.6±30
	clDice	**6.2±6.2**	**127±15**	**4.0±3.0**	**38.7±8.0**
	Warping	11.4±6.5	190±20	**4.5±2.8**	**43.6±12**
	SkelRecall	9.9±3.4	**160±12**	5.2±4.0	69.2±10
	cbDice	9.6±4.7	**159±15**	4.9±2.9	**54.8±8.3**
Pseudo-labels	CEDice	12.3±1.5	126±26	12.0±1.5	83.9±15
	RWLoss	**11.4±1.4**	**114.7±25**	**11.11±2.0**	96.5±23
	TopoLoss	**10.4±1.1**	**118±15**	**10.1±1.4**	110.3±33
	TOPO	36.1±5.1	225.4±37	31.1±2.8	132±41
	clDice	**1.8±0.4**	**32.7±6.1**	**4.3±0.4**	**61.6±12**
	Warping	13.6±1.7	127±39	13.3±2.3	95.5±20
	SkelRecall	**10.4±1.6**	**113.6±25**	**10.4±1.7**	85.8±21
	cbDice	13.1±3.5	141.4±33	14.7±2.5	123±28
Noisy labels	CEDice	6.2±1.1	175±14	5.1±0.6	18.7±5.5
	RWLoss	17.9±3.7	427.5±26	11.9±1.0	**17.9±3.1**
	TopoLoss	7.2±2.5	205±12	6.6±0.4	**14.9±1.3**
	TOPO	24.1±9.4	**157±32**	10.9±4.8	140±22
	clDice	**5.0±0.5**	**126.5±13**	6.8±0.5	48.2±16
	Warping	8.5±1.7	285±19	9.5±0.6	**18.0±1.8**
	SkelRecall	**3.8±1.6**	**113±14**	**1.5±0.4**	44.6±7.9
	cbDice	10.5±1.4	276±14	9.7±0.6	27.5±6.6

		β_0 error ID	β_0 error OOD	β_1 error ID	β_1 error OOD
D.A. (*RandHue*)	CEDice	1.9±2.9	69.7±15	2.0±5.0	53.3±12
	RWLoss	3.3±1.0	**62.8±5.8**	2.9±1.9	**49.8±3.6**
	TopoLoss	2.9±4.2	85.3±20	3.3±7.8	80.1±19
	TOPO	19.7±2.3	97.2±11	40.6±1.4	**49.9±11**
	clDice	**0.6±0.2**	**16.5±3.1**	**1.0±0.1**	**15.1±2.6**
	Warping	2.5±1.3	**57.1±5.8**	1.9±1.3	**40.0±3.7**
	SkelRecall	2.0±2.8	65.5±8.7	2.0±4.8	62.7±9.3
	cbDice	2.7±1.2	90.3±18	3.3±4.7	83.2±18
Pseudo + Self. dist.	CEDice	3.8±0.4	68.4±14	4.0±0.5	47.4±13
	RWLoss	4.4±0.4	73.0±13	4.3±0.5	**43.9±8.6**
	TopoLoss	7.6±1.4	**34.9±6.6**	**3.6±0.4**	**31.7±6.7**
	TOPO	64.5±9.9	131.3±19	73.1±6.5	57.5±41
	clDice	**2.1±0.4**	**32.6±2.4**	**2.7±0.3**	55.9±10
	Warping	4.6±0.5	**66.3±5.6**	4.8±0.5	**42.7±7.4**
	SkelRecall	4.7±0.4	81.5±17	5.4±0.6	61.8±15
	cbDice	4.5±0.6	78.4±14	4.8±0.5	61.4±19
Noisy + Self. dist.	CEDice	2.4±0.7	114±9.4	2.8±0.4	13.5±1.7
	RWLoss	13.5±2.0	289±23	16.6±1.5	17.9±0.8
	TopoLoss	5.6±1.7	113±11	8.3±1.3	15.7±0.7
	TOPO	67.2±46	118±36	9.3±6.3	81.0±27
	clDice	**0.9±0.3**	**62.8±9.6**	**1.6±0.2**	22.5±3.0
	Warping	2.8±0.3	153±11	4.0±0.4	16.2±1.8
	SkelRecall	**1.4±0.5**	**62.8±8.5**	**1.0±0.1**	36.8±7.7
	cbDice	4.7±1.0	163±13	6.3±1.2	23.6±28

Table 3: Average performance on TopoMortar test set. Top: training setup as in Table 2 but on a small training set or with *RandHue* data augmentation. Center: Pseudo-labels without, and with self-distillation. Bottom: Noisy labels, without and with self-distillation. **Bold**: Significantly lower than CEDice. Red: Smallest average. Blue: Second smallest.

The models' performance decreased considerably after introducing the aforementioned dataset challenges, with an average Dice coefficient in the ID test set of 0.90, 0.86, and 0.68

in the scarce training data, pseudo-label, and noisy label setups, respectively (Appendix L). On both ID and OOD test sets, clDice achieved the lowest Betti errors when training on the small training set and when training on pseudo-labels, whereas Skeleton Recall loss was generally superior with noisy labels (Table 3 (left)). The second-best topology loss function depended on the experimental setup and the Betti error (Table 3 (left), blue). In the OOD test set, according to the β_0 error, TopoLoss was the second best in the small training set setup, Skeleton Recall in the pseudo-labels, and clDice in the noisy-labels experiment. According to the β_1 error, Warping loss in the small training set, CEDice in the pseudo-labels, and RWLoss in the noisy-labels experiment.

4.4 Topology losses with data augmentation and self-distillation

We studied the impact on topology accuracy of two simple methods for tackling dataset challenges. First, we accounted for the presence of OOD images with a simple *data augmentation* method that increased the colors' diversity in the images and that we applied with a probability of 50%. This data augmentation, which we refer to as *RandHue*, converted the images to the HSV color space; randomly chose the same hue for all pixels; randomly shifted the saturation and value; and converted the image back to RGB (see examples in Appendix M). Second, we accounted for labels being pseudo-labels and noisy labels by training with *self-distillation* [19]—a strategy known to be advantageous with those types of labels. We employed self-distillation due to its simplicity and because it incorporated no extra hyper-parameters. To keep the total iterations to 12,000 as in all our experiments, we trained the models for 4,000 iterations, generated soft labels for the training set, trained on those labels for another 4,000 iterations, and repeated the process one more time.

The segmentations and, particularly, their topology accuracy were generally better than in the previous experiment where no dataset challenge was directly tackled. clDice also achieved the best segmentations in most cases, and Skeleton Recall also outperformed in the presence of noisy labels. As in the previous experiment, the second-best topology loss function depended on the specific experimental setup. Applying RandHue improved the Dice coefficients in the OOD images and decreased the Betti errors significantly in both the ID and OOD test set (Table 2 vs. Table 3 (top-right)). The decrease in the β_0 error occurred across all OOD groups, whereas the decrease in the β_1 error occurred only in "angles", "colors", and "shadows" (see Appendix N). When training on pseudo-labels and noisy labels, self-distillation improved Dice coefficients and topology accuracy.

5 Discussion

We presented TopoMortar, a dataset specifically created to study the effectiveness of topology-focused image segmentation methods. We showed that existing datasets exhibit challenges that were not previously considered and that influence topology accuracy. We compared eight loss functions on TopoMortar on a setup without dataset challenges and then studied model robustness in the presence of those challenges. Finally, we investigated the extent to which simple data augmentation and self-distillation can increase topology accuracy.

We evaluated topology loss functions on different datasets, following the standard experimental approach. Additionally, we tackled the challenges of data scarcity and noisy labels in DRIVE and CrackTree datasets with a larger dataset and with a method to learn from noisy labels, respectively. Our experiments revealed two key points. First, no topology loss

function was the best across all settings, which contrasts with topology loss function studies where the proposed loss function always outperforms the others. Second, by tackling dataset challenges, not only topology accuracy improves but also the relative advantageousness of the topology loss functions changes. Importantly, this experiment **does not reveal when and why** specific topology loss functions are advantageous, as comparing across datasets entangles dataset tasks and dataset challenges. For instance, cbDice and Warping loss achieved higher topology accuracy than CEDice in the DRIVE dataset while, on CrackTree, they did not surpass CEDice. This may indicate that cbDice and Warping loss lead to models robust against scarce training data but not against noisy labels; or that cbDice and Warping loss are particularly well suited for blood vessel segmentation; or both.

We evaluated the topology loss functions on TopoMortar **eliminating dataset challenges** by ensuring sufficient training data, training time, accurate labels, ID test-set images, strong data augmentation, and a state-of-the-art deep learning model. This scenario was either assumed or not discussed in previous works. In this challenging setup, only clDice and Skeleton Recall achieved β_1 errors smaller and significantly different than CEDice, with clDice also achieving smaller and significantly different β_0 errors, demonstrating the potential of skeletonization-based topology loss functions.

We investigated model **robustness against dataset challenges** after optimizing topology loss functions on TopoMortar. In contrast to experiments on other datasets, TopoMortar allows fixing the dataset task (*i.e.*, segmenting mortar in red brick walls), permitting to study the effect of each dataset challenge, individually, on the potential advantageousness of topology losses. We observed 1) that clDice was generally the best-performing loss, 2) that Skeleton Recall worked best specifically under the presence of noisy labels, and 3) that the performance of the other losses varied depending on the dataset challenge. These results, in line with the other experiments, indicate that clDice is truly effective at enhancing topology accuracy. The outperformance of Skeleton Recall over clDice on noisy labels can be explained by its emphasis on the foreground region (true positives, false negatives), as, on TopoMortar's noisy labels, the foreground corresponds to a thicker and more accurate skeleton than what clDice produces. Thus, it may be that with a different type of noisy labels [1] Skeleton Recall performs differently.

We also studied the impact of data augmentation and self-distillation on topology accuracy. These simple and well-established strategies improved the baseline CEDice when training on a large training set with accurate labels and when optimizing on noisy and pseudo-labels (Tables 2 and 3), making CEDice outperform the majority of topology loss functions. Although it is unsurprising that data augmentation, self-distillation, and other methods [6] improve performance, limited research has investigated to what extent they can increase topology accuracy, or even if they can make standard models trained on CEDice surpass topology loss functions. Considering that topology loss functions are generally computationally expensive CPU- and GPU-wise (Appendix O), our results demonstrate that focusing on improving regular accuracy by utilizing methods that account for dataset challenges can be a resource-friendly alternative to topology loss functions to increase topology accuracy. Moreover, combining such methods with topology loss functions further improved topology accuracy in most cases, especially in the OOD test-set images (Table 3).

TopoMortar was designed to be simple to prevent methods from exploiting dataset particularities to increase topology accuracy. Despite mortar's relatively simple topology, topology accuracy on TopoMortar has a very high correlation with the topology accuracy on CREMI, DRIVE, FIVES, and CrackTree datasets (Appendix P), demonstrating the **generalizability** of results across biological, non-biological datasets, and structures with different topology.

6 Conclusion

Previous benchmarks on existing datasets have not allowed to completely understand whether methods improve topology accuracy merely by focusing on the dataset characteristics (thus, disregarding topology), or by learning data's topological properties. In contrast, our TopoMortar dataset permits to study this research question by eliminating confounding factors and by enabling the investigation of model robustness against various dataset challenges. clDice generally achieved the most topologically accurate segmentations while Skeleton Recall performed best on noisy labels. Additionally, data augmentation and self-distillation helped in improving topology accuracy, even when employed jointly with topology loss functions.

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